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Self-hosted GLPI ticketing system integrated with Active Directory via encrypted LDAPS, extending the active-directory-lab project

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📖 README

Helpdesk Ticketing Lab: TICKET01

A self-hosted GLPI ticketing system built in VirtualBox and integrated with the existing lab.local Active Directory environment, simulating a small company's IT service desk.

Built over 6–7 September 2026, extending [active-directory-lab](#) — TICKET01 joins DC01 and CLIENT01 on the same isolated network, authenticating against the same domain.

Project Goal

To add a service-desk layer on top of an existing enterprise IT environment: deploy an open-source ITSM tool, integrate it with Active Director for real user authentication (upgraded to encrypted LDAPS with a real internal CA), and populate it with a realistic ticket history drawn from genuine issues hit while building the infrastructure itself.

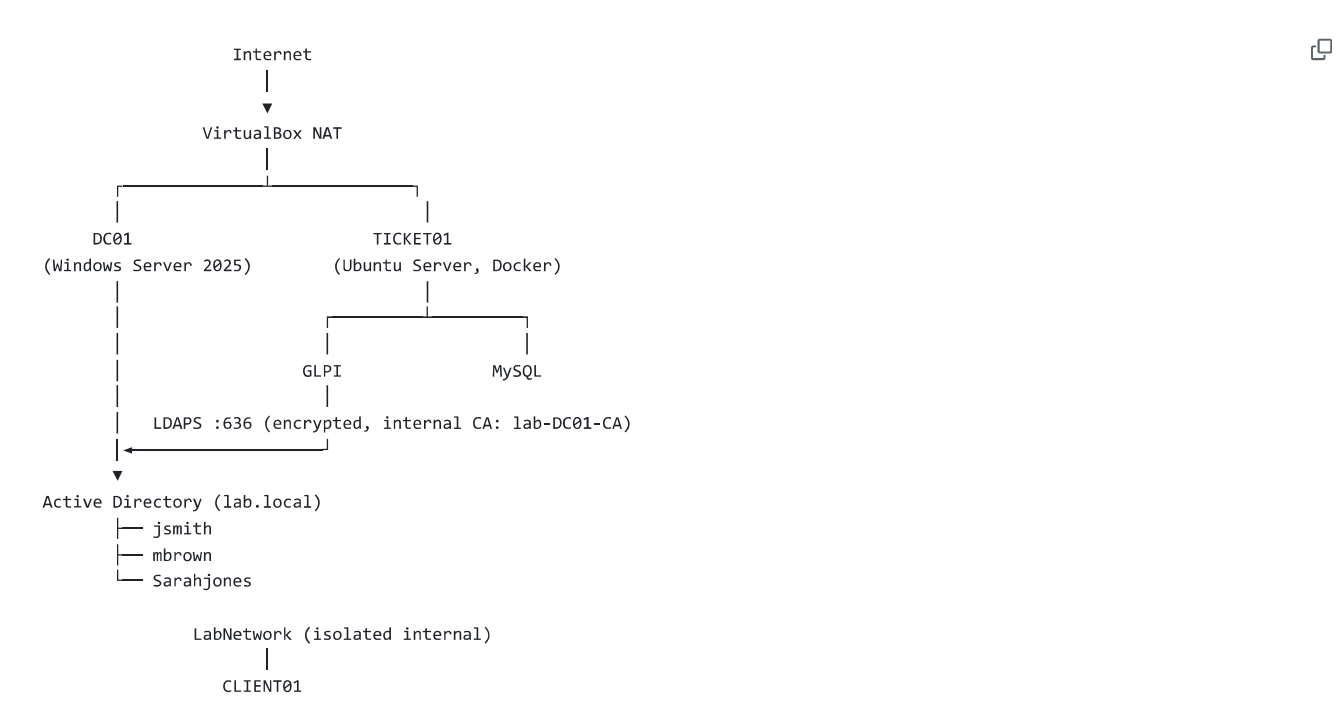
Environment Overview

Component	Details
Hypervisor	Oracle VirtualBox 7.2.16
Domain Controller	dc01 , Windows Server 2025 (see active-directory-lab)
Ticketing Server	TICKET01 , Ubuntu Server 26.04.1 LTS, minimized install
Ticketing Software	GLPI (official glpi/glpi Docker image) + MySQL, via Docker Compose
Domain	lab.local
Networking	Dual-homed: NAT (management/internet) + isolated LabNetwork (same as DC01/CLIENT01)
TICKET01 LabNetwork IP	192.168.1.30 (static, via netplan)
DNS	Points to DC01 (192.168.1.10)
Remote access	SSH from Windows PowerShell, via NAT port-forward (host 2222 → guest 22)
GLPI web access	Browser via NAT port-forward (host 8080 → guest 80)

Component	Details
LDAP / LDAPS	lab.local , authenticated against DC01, port 636 (encrypted)

TICKET01 sits on the same isolated internal network as the existing domain controller and client, reflecting how a real internal service-desk server would sit inside a company's LAN rather than being a bolt-on SaaS tool. It is dual-homed — one adapter for host management (NAT), i.e. second dedicated purely to talking to the domain (LabNetwork) — the same pattern real servers use to separate management traffic from production traffic.

Architecture Overview



GLPI (on TICKET01) authenticates against DC01 over LDAPS rather than plain LDAP — see Troubleshooting Log, Issues 6–11, for the full path from a plaintext workaround to a properly trusted, hostname-verified, persistent TLS configuration.

Design Rationale

GLPI was chosen over other open-source options (osTicket, Zammad) specifically because it combines ticket management with native LDAP/Active Directory authentication — allowing genuine integration with the existing domain rather than a standalone tool with its own separate user base. This mirrors how commercial ITSM platforms commonly referenced in UK IT support job listings (ServiceNow, Freshservice) integrate with corporate directory services.

Security rationale:

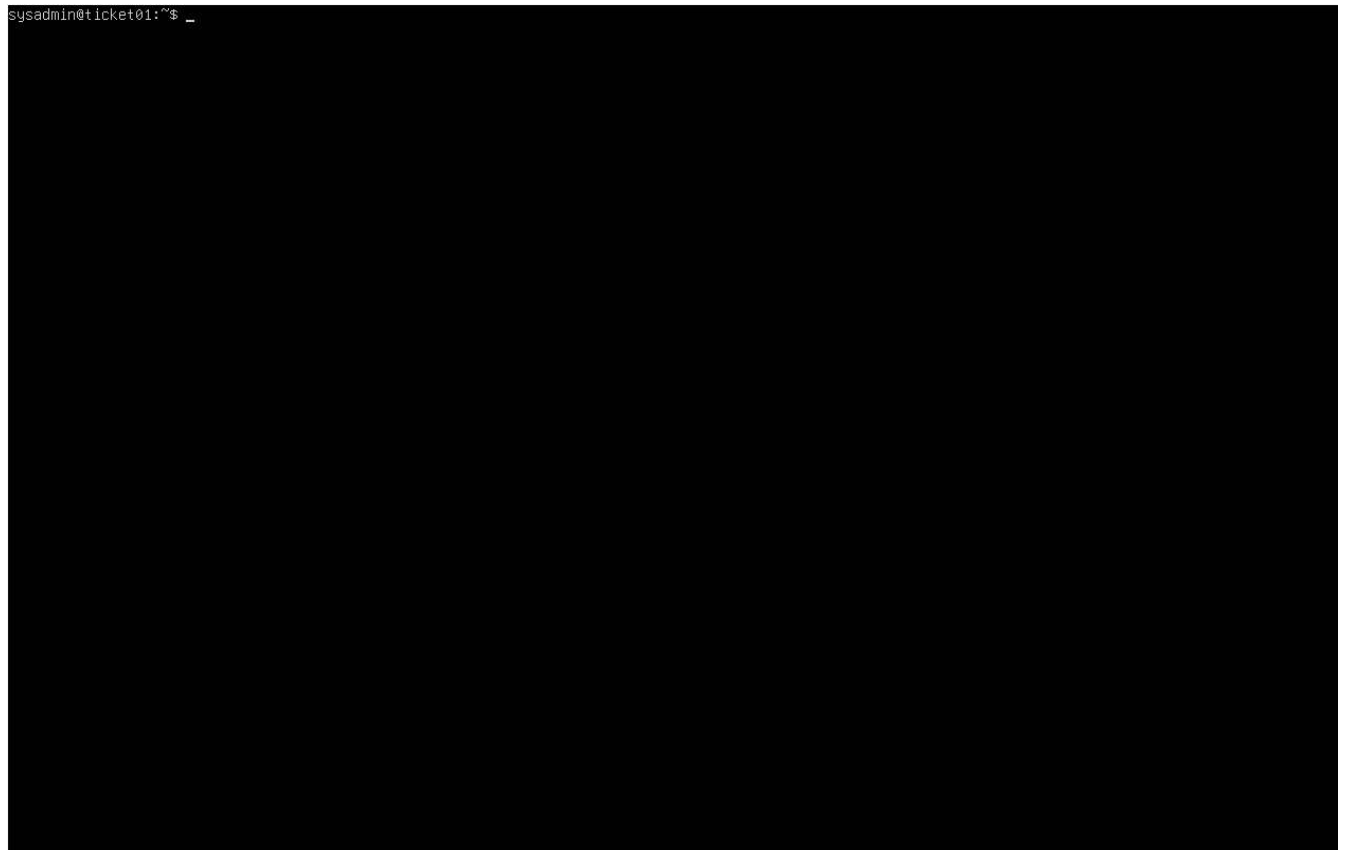
- A dedicated, least-privilege service account (`svc-glpi-ldap`) binds to LDAP — not a personal admin account
- No real credentials are committed to this repo — `.env` and the real LDAPS certificate/config are excluded via `.gitignore` ; `.env.example` documents the required variables without real values
- LDAPS (port 636, internal CA) replaces the original plain-LDAP-with-relaxed-signing workaround once the production-correct fix was built (see Issue 6 and Section 10 below)
- GLPI's built-in demo data was disabled, so no placeholder assets/tickets ship with the deployed instance

What Was Built

1. TICKET01 VM provisioning

Ubuntu Server 26.04.1 LTS installed via VirtualBox unattended installation (minimized install), initially on NAT networking to allow package downloads.

```
sysadmin@ticket01:~$ _
```



2. Remote administration via SSH

OpenSSH installed during setup; administered remotely from Windows PowerShell via a NAT port-forwarding rule (host 2222 → guest 22) rather than the local VirtualBox console.

3. Docker & GLPI deployment

Docker Engine and Docker Compose installed via Ubuntu's official repository. GLPI deployed via the official `glpi/glpi` image, paired with a MySQL container, defined in `docker-compose.yml` .

```

sysadmin@ticket01: ~/glpi-dc
sysadmin@ticket01:~/glpi-docker$ docker compose up -d
unable to get image 'glpi/glpi:latest': permission denied while trying to connect to the Docker daemon socket at unix:///var/run/d
ocker.sock: Get "http://%2Fvar%2Frun%2Fdocker.sock/v1.51/images/glpi/glpi:latest/json": dial unix /var/run/docker.sock: connect: p
ermission denied
sysadmin@ticket01:~/glpi-docker$ sudo usermod -aG docker $USER
sysadmin@ticket01:~/glpi-docker$ exit
logout
Connection to 127.0.0.1 closed.
PS C:\WINDOWS\System32> ssh -p 2222 sysadmin@127.0.0.1
sysadmin@127.0.0.1's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

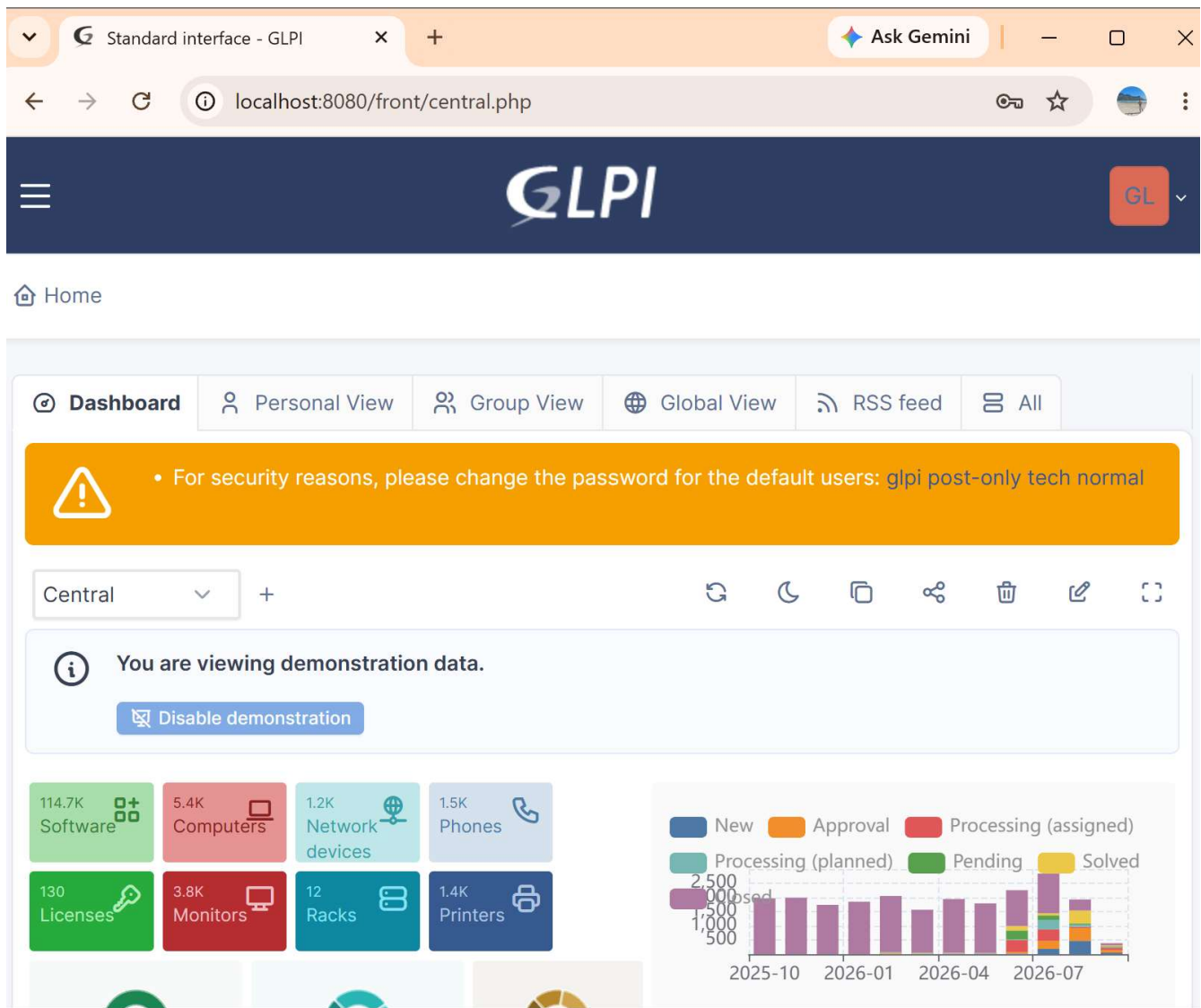
This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Sun Sep  6 13:43:21 2026 from 10.0.2.2
sysadmin@ticket01:~$ cd ~/glpi-docker
sysadmin@ticket01:~/glpi-docker$ docker compose up -d
[+] Running 43/43
  ✓glpi Pulled                               120.4s
  ✓db Pulled                                 110.6s
[+] Running 5/5
  ✓Network glpi_default      Created          0.3s
  ✓Volume glpi_glpi_data     Created          0.0s
  ✓Volume glpi_db_data       Created          0.0s
  ✓Container glpi-db-1       Started         2.1s
  ✓Container glpi-glpi-1     Started         2.2s
sysadmin@ticket01:~/glpi-docker$

```

4. First GLPI access

A second NAT port-forwarding rule (host 8080 → guest 80) exposed GLPI's web interface. GLPI auto-installed its database schema on first run skipping straight to a working login screen.



5. Dual-homed networking for LabNetwork integration

A second adapter (`enp0s8`) was added to TICKET01 on the isolated `LabNetwork` , given a static IP (`192.168.1.30/24`) via `netplan`, continuing the addressing scheme from the AD lab (DC01 `.10` , CLIENT01 `.20`). The original NAT adapter was left untouched, so remote management was never lost.

```
sysadmin@ticket01: ~  
link/ether 7a:be:68:f3:15:bd ff:ff:ff:ff:ff:ff  
inet 172.18.0.1/16 brd 172.18.255.255 scope global br-fc5dffcbfb00  
valid_lft forever preferred_lft forever  
inet6 fe80::70be:68ff:fe80:f315/64 scope link proto kernel ll  
valid_lft forever preferred_lft forever  
6: vetha5048ce@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-fc5dffcbfb00 state UP group default  
link/ether 96:7d:f4:8b:e2:8b ff:ff:ff:ff:ff:ff link-netnsid 0  
inet6 fe80::947d:f4ff:fe8b:e28b/64 scope link proto kernel ll  
valid_lft forever preferred_lft forever  
7: vetha88b3b6@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-fc5dffcbfb00 state UP group default  
link/ether e2:9e:ed:8e:14:23 brd ff:ff:ff:ff:ff:ff link-netnsid 1  
inet6 fe80::e09e:edff:fe8e:1423/64 scope link proto kernel ll  
valid_lft forever preferred_lft forever  
sysadmin@ticket01:~$ ping -c 4 192.168.1.10  
-bash: ping: command not found  
sysadmin@ticket01:~$ ping -c 4 192.168.1.10  
-bash: ping: command not found  
sysadmin@ticket01:~$ sudo apt install -y iputils-ping  
The following packages were automatically installed and are no longer required:  
  linux-headers-7.0.0-30      linux-image-unsigned-7.0.0-30-generic  linux-modules-7.0.0-30-generic  linux-tools-7.0.0-30-generic  
  linux-headers-7.0.0-30-generic  linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30  
Use 'sudo apt autoremove' to remove them.  
  
Installing:  
  iputils-ping  
  
Summary:  
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 0  
Download size: 47.1 kB  
Space needed: 198 kB / 2371 MB available  
  
Get:1 http://gb.archive.ubuntu.com/ubuntu resolute/main amd64 iputils-ping amd64 3:20250605-1ubuntu1 [47.1 kB]  
Fetched 47.1 kB in 0s (216 kB/s)  
debconf: unable to initialize frontend: Dialog  
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79, <STDIN> line 1.)  
debconf: falling back to frontend: Readline  
Selecting previously unselected package iputils-ping.  
(Reading database ... 128609 files and directories currently installed.)  
Preparing to unpack .../iputils-ping_3:20250605-1ubuntu1_amd64.deb ...  
Unpacking iputils-ping (3:20250605-1ubuntu1) ...  
Setting up iputils-ping (3:20250605-1ubuntu1) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
sysadmin@ticket01:~$ ping -c 4 192.168.1.10  
PING 192.168.1.10 (192.168.1.10) 56(84) bytes of data.  
64 bytes from 192.168.1.10: icmp_seq=1 ttl=128 time=2.38 ms  
64 bytes from 192.168.1.10: icmp_seq=2 ttl=128 time=3.86 ms  
64 bytes from 192.168.1.10: icmp_seq=3 ttl=128 time=2.25 ms  
64 bytes from 192.168.1.10: icmp_seq=4 ttl=128 time=3.93 ms  
  
--- 192.168.1.10 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3007ms  
rtt min/avg/max/mdev = 2.247/3.103/3.932/0.791 ms  
sysadmin@ticket01:~$
```

6. LDAP integration with lab.local

A dedicated least-privilege service account (svc-glpi-ldap) was created in a new Service-Accounts OU on DC01 for this integration. GLPI's LDAP directory was configured to bind using this account — initially over plain LDAP, which failed due to DC01's default signing enforcement (Issue 6), then resolved. jsmith, Sarah Jones, and Mike Brown were imported from their respective OUs as real GLPI users.

7. Ticket categories & templates

Five categories created matching real IT support taxonomy: Hardware, Software, Network, Account & Access, and Active Directory.

8. Realistic ticket population

Nine tickets created across all three imported AD users and all five categories, mixing genuine technical incidents drawn from this project's own troubleshooting log with ordinary day-to-day helpdesk volume:

#	Title	Requester	Category	Status
1	New server (TICKET01) failed to boot after setup	jsmith	Active Directory	Closed
2	Can't get GLPI to authenticate against AD, login keeps failing	jsmith	Active Directory	Closed
3	New starter laptop keeps setting up Windows from scratch every time I turn it on	mbrown	Active Directory	Processing
4	My wallpaper policy isn't applying even though IT said they fixed it	Sarahjones	Active Directory	Processing
5	Locked out of my account after too many password attempts	mbrown	Account & Access	Closed
6	Printer in Sales not responding	Sarahjones	Hardware	Processing
7	Can't connect to the office Wi-Fi from my laptop	jsmith	Network	Processing
8	Need Excel installed on my new PC	mbrown	Software	Processing
9	New starter needs a Finance-Department account setup	Sarahjones	Account & Access	Processing

Three tickets (the RCU stall, the LDAP signing failure, and an account lockout) were resolved with real, specific solution text and closed — the remainder left in "Processing" status, reflecting a realistic snapshot of an active service desk rather than a fully-wrapped demo.

Tickets - GLPI

Ask Gemini

localhost:8080/front/ticket.php

Home / Assistance / Tickets + Add Templates Global Kanban

Search... Super-Admin Root entity (tree structure)

1.5K Tickets

240 Incoming tickets

67 Pending tickets

308 Assigned tickets

4 Planned tickets

78 Solved tickets

14.6K Closed tickets

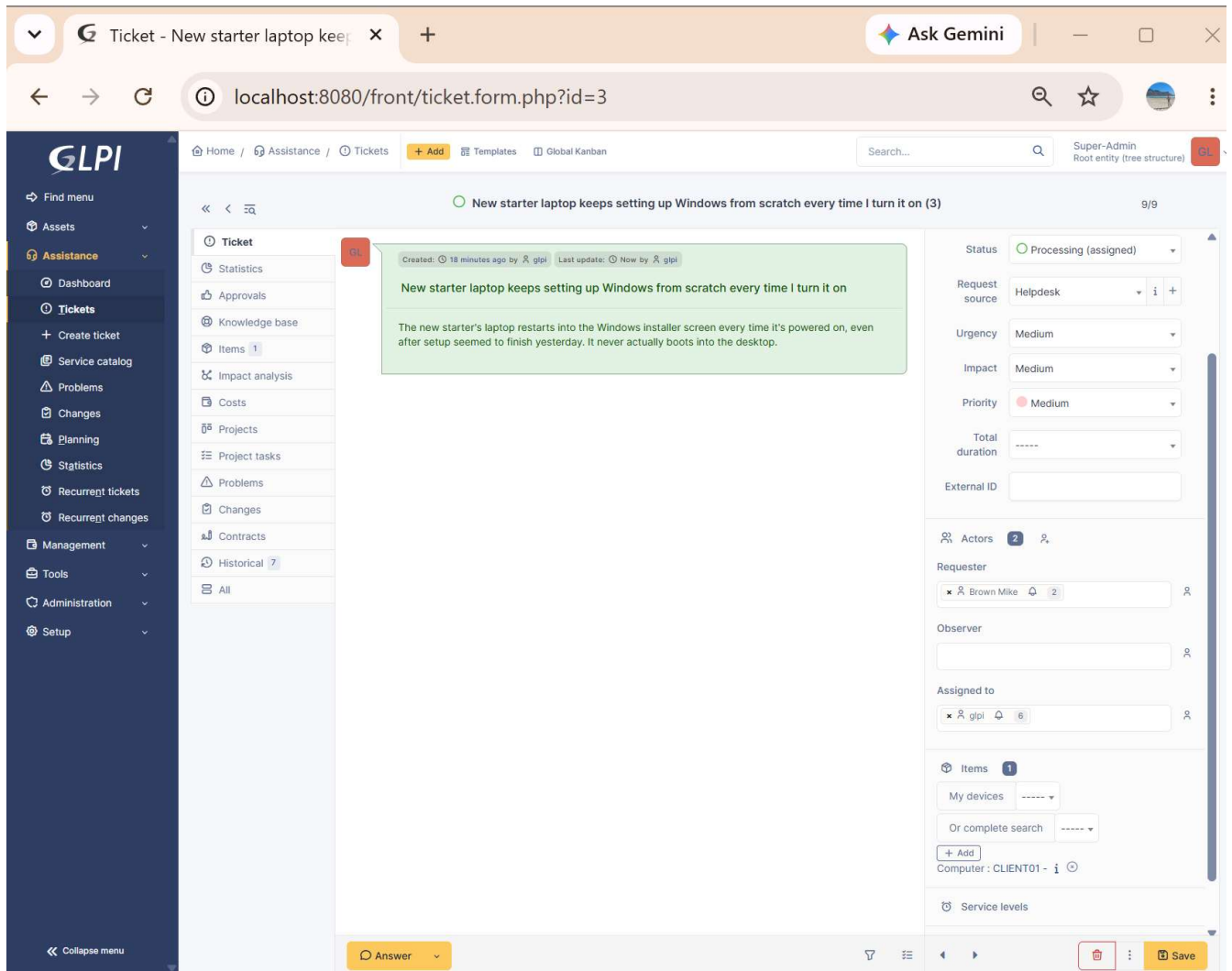
Search Sorted by Last update

ID	TITLE	STATUS	LAST UPDATE	OPENING DATE	PRIORITY	REQUESTER - REQUESTER	ASSIGNED TO - TECHNICIAN	CATEGORY	TIME TO RESOLVE
9	New starter needs a Finance-Department account setup	Processing (assigned)	2026-09-07 10:55	2026-09-07 12:00	Medium	Jones Sarah	glpi	Account & Access	
8	Need Excel installed on my new PC	Processing (assigned)	2026-09-07 10:54	2026-09-07 12:00	Low	Brown Mike	glpi	Software	
7	Can't connect to the office Wi-Fi from my laptop	Processing (assigned)	2026-09-07 10:54	2026-09-07 12:00	Low	Smith John	glpi	Network	
6	Printer in Sales not responding	Processing (assigned)	2026-09-07 10:53	2026-09-07 12:00	Medium	Jones Sarah	glpi	Hardware	
5	Locked out of my account after too many password attempts	Processing (assigned)	2026-09-07 10:52	2026-09-07 12:00	Medium	Brown Mike	glpi	Account & Access	
4	My wallpaper policy isn't applying even though IT said they fixed it	Processing (assigned)	2026-09-07 10:52	2026-09-07 12:00	Low	Jones Sarah	glpi	Active Directory	
3	New starter laptop keeps setting up Windows from scratch every time I turn it on	Processing (assigned)	2026-09-07 10:51	2026-09-07 12:00	Medium	Brown Mike	glpi	Active Directory	
2	Can't get GLPI to authenticate against AD, login keeps failing	Processing (assigned)	2026-09-07 10:50	2026-09-07 12:00	Medium	Smith John	glpi	Active Directory	
1	New server (TICKET01) failed to boot after setup	Processing (assigned)	2026-09-07 10:44	2026-09-07 12:00	High	Smith John	glpi	Active Directory	

20 rows / page Showing 1 to 9 of 9 rows

9. Asset inventory

GLPI's asset side was populated with the three real lab machines (DC01, CLIENT01, TICKET01). Ticket #3 was linked to its real subject asset (CLIENT01) via the ticket's Items tab, demonstrating GLPI's asset-ticket relationship rather than tickets existing as disconnected text records.



10. LDAPS: replacing the signing workaround with real encryption

The original LDAP integration (Section 6) worked over plain LDAP on port 389, only after relaxing DC01's signing requirements — a deliberate lab simplification, not a production-correct fix. This stretch goal replaced it with genuine LDAPS (port 636) backed by a real internal Certificate Authority:

- Installed AD CS on DC01 as an Enterprise Root CA (`1ab-dc01-ca`); DC01 auto-enrolled its own Domain Controller certificate
- Exported the CA certificate and trusted it inside the `glpi` container via `TLS_CACERT`
- Diagnosed and fixed a TLS hostname-verification mismatch (GLPI was connecting by IP; the certificate's subject is the hostname) — see Issues 7–10
- Made the entire fix persistent through `docker-compose.yml` (a bind-mounted `ldap-config` volume plus an `extra_hosts` entry), verified against a **full container rebuild**, not just a restart

GLPI now authenticates against `1ab.1oca1` over genuinely encrypted LDAPS, with the entire trust and hostname-resolution configuration defined declaratively rather than living only inside a running container's memory.

Full step-by-step screenshots for each stage are in the `screenshots` folder (48 in total, numbered in build order).

Setup & Deployment

Prerequisites

- Oracle VirtualBox 7.2.16+
- An existing Active Directory domain reachable from TICKET01 (see [active-directory-lab](#) for building one from scratch)
- Ubuntu Server 26.04.1 LTS ISO
- Docker Engine & Docker Compose

Installation Steps

1. **Provision the VM**, dual-homed: one adapter on NAT (management), one on the same internal network as your domain controller.

2. **Install Docker**, then clone this repo onto the VM:

```
git clone https://github.com/JackBraun01/helpdesk-ticketing-lab.git
cd helpdesk-ticketing-lab
```



3. **Configure environment variables** — copy the template and fill in real values (never commit the result):

```
cp .env.example .env
nano .env
```



4. **Deploy GLPI + MySQL**:

```
docker compose up -d
```



5. **Configure LDAP** in GLPI's web UI (Setup → Authentication → LDAP directories), pointing at your domain controller with a dedicated service account.

6. **Upgrade to LDAPS** (recommended over plain LDAP):

- Stand up an Enterprise Root CA on your domain controller (AD CS role) and export its certificate
- Place the exported `.pem` and a matching `ldap.conf` (with a `TLS_CACERT` line) into `./ldap-config/` — this folder is bind-mounted into the container automatically via `docker-compose.yml`
- In GLPI's LDAP directory settings, set the Server field to `ldaps://<your-dc-hostname>` and Port to `636`

Verification

```
docker compose ps
nc -zv <dc-hostname> 636
docker exec -it glpi-glpi-1 getent hosts <dc-hostname>
docker compose down && docker compose up -d # confirm config survives a full rebuild
```



Then re-run GLPI's built-in LDAP connection test (Setup → Authentication → LDAP directories → Test) and confirm all five stages pass.

Troubleshooting Log

Real issues encountered and resolved during the build, included deliberately — diagnosing and fixing genuine problems, not just following steps that work first time, is what actually demonstrates capability.

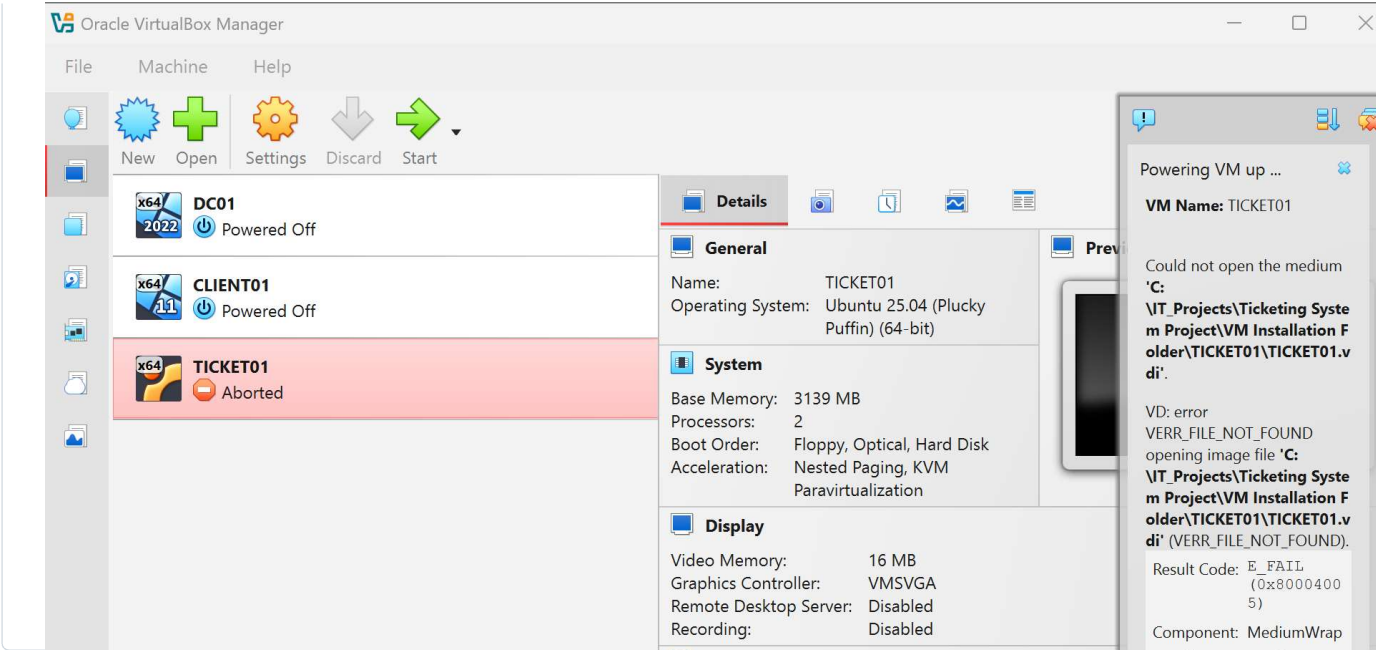
#	Issue	Root Cause	Resolution
1	TICKET01 VM registered in VirtualBox but failed to boot with <code>VERR_FILE_NOT_FOUND</code> , "Could not open the medium"	The virtual hard disk (<code>.vdi</code>) file never actually finished being created during VM creation, despite the VM's configuration files registering successfully	Removed the broken VM registration, recreated the VM, and explicitly inspected the "Specify virtual hard disk" panel before clicking Finish
2	TICKET01 boot hung indefinitely at "Loading essential drivers" during install, with no progress after 10+ minutes	Kernel-level RCU stall caused by a timing/scheduling desync between VirtualBox 7.2.x and multi-core Linux guests — a known, currently open issue on VirtualBox's tracker	Reduced the VM's allocated CPU count from 2 to 1. Before concluding this was the cause, ruled out a corrupted ISO (SHA256 checksum matched Canonical's published hash exactly) and host resource contention
3	<code>docker compose up -d</code> failed with permission denied while trying to connect to the Docker daemon socket	A freshly created non-root user isn't automatically granted access to Docker's restricted Unix socket — requires explicit <code>docker</code> group membership	Added the user to the <code>docker</code> group (<code>sudo usermod -aG docker \$USER</code>), then fully logged out and reconnected, since group membership only applies to a new session
4	Browser could not reach GLPI (<code>ERR_CONNECTION_REFUSED</code>), despite the container working correctly internally	Only an SSH (port 22) NAT forwarding rule existed; port 80 had no rule, so the host browser had no route in	Added a second NAT port-forwarding rule (host <code>8080</code> → guest <code>80</code>)

#	Issue	Root Cause	Resolution
5	Second adapter's static IP failed to apply; netplan apply errored with unknown key 'enp0s8'	The netplan YAML was accidentally overwritten rather than appended to while editing, leaving incorrect indentation	Rebuilt the netplan file from scratch with correct nesting for both adapters, verifying structure with cat -A before reapplying
6	GLPI's LDAP test failed with Strong(er) authentication required(8) at Bind connection	Windows Server 2025 enforces LDAP signing by default, rejecting unsigned binds over plain LDAP (port 389)	Relaxed LDAP signing requirements via Group Policy as a deliberate, documented lab simplification — flagged at the time as the reason a production-correct LDAPS fix (Section 10) was needed
7	After switching to LDAPS, GLPI's test passed TCP/Base DN/LDAP URI but failed at Bind with Can't contact LDAP server(-1)	The TLS handshake was rejected because the glpi container's isolated trust store had never been told to trust the new internal CA	Exported the CA certificate and registered it via TLS_CACERT in the container's ldap.conf
8	Certificate text pasted into the SSH session landed directly at the bash prompt instead of inside a file	nano hadn't been opened first — pasting multi-line text straight at a shell prompt causes bash to interpret each line as a command	Opened nano lab-DC01-CA.pem first, then pasted the still-available clipboard content into the editor
9	mkdir /etc/ldap inside the container failed with Permission denied; running docker commands from inside the container failed with command not found	docker exec connects as the low-privilege www-data user by default; docker itself is a host-level CLI tool not installed inside the container	Re-ran container commands with docker exec -u root from the TICKET01 host shell
10	After adding TLS_CACERT, the identical bind failure persisted	GLPI connected via DC01's IP (ldaps://192.168.1.10), but the certificate's subject is CN=DC01.lab.local — a TLS hostname-verification mismatch. The container also couldn't resolve the hostname at all (separate DNS from the host VM)	Added a direct /etc/hosts entry inside the container and switched GLPI's Server field to the hostname. All five test stages passed
11	A cat > file << 'EOF' heredoc paste corrupted docker-compose.yml with duplicated, malformed content	The heredoc's closing EOF merged with a second, duplicate paste on the same line with no line break, so bash never recognized the terminator and kept absorbing input	Deleted the file and re-ran the exact same heredoc as a single, uninterrupted paste, confirming with cat before proceeding

Evidence

Screenshots are split into two numbered sequences: `doc-NN` (standard build steps) and `ts-NN` (errors/troubleshooting and their fixes), in the `Screenshots` and `Troubleshooting` folders respectively.

Issue 1: Missing virtual disk file



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Contributors 1

 JackBraun01

Languages

PowerShell 100%